20 Processing, Cryopreserving, and Controlling the Quality of HSC

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20.1 Assessment of HSC by Measuring CD34 and the Presence of Other Cell Subsets

The efficiency of an autologous, as well as an allogeneic, HSC graft is mainly determined by the number of $CD34^+$ cells present. The dose of transplanted $CD34^+$ cells per kg body weight (bw) determines the kinetics of the neutrophil and platelet engraftment after auto-HCT (Weaver et al. 1995). The measurement of $CD34^+$ cells by flow cytometry is, therefore, an important method to assess the graft quantity.

The minimal number of $CD34^+$ cells for an autologous transplant is $\geq 2.0 \times 10^6$ $CD34^+$ cells/kg bw (Mohty et al. 2014). Transplants below this threshold should only be used in cases where no additional stem cell collection is feasible, and there is a vital indication for the autologous stem cell transplantation. Most transplant centres regard a cell dose of $2.5–5 \times 10^6$ $CD34^+$ cells/kg bw as optimal, based on the published clinical data (Duong et al. 2014; Perez-Simon et al. 1999; Giralt et al. 2014). For an allo-HCT, a cell dose of $\geq 4.0 \times 10^6$ $CD34^+$ cells/kg bw is regarded as adequate in most cases. If further processing (i.e., cryopreservation, cell depletion, or purging) is planned, higher doses of $5–8 \times 10^6$ $CD34^+$ cells/kg bw could be necessary.

In the autologous setting, it has been speculated that the quality of $CD34^+$ cells from poor mobilizers may be inferior. However, studies have found that the proportions of primitive and quiescent $CD34^+$ subsets were comparable across mobilization groups (Jiang et al. 2012), and leukocyte and platelet recovery after transplantation was not different (Wuchter et al. 2010). The application of plerixafor in order to overcome insufficient HSC mobilization not only increases the number of $CD34^+$ cells but also the proportion of more primitive HSC subsets, the absolute lymphocyte count, and the numbers of lymphocytes in various subsets ($CD19^+$ cells, $CD3^+$ cells, T-cells, and NK-cells) in the autograft (Fruehauf et al. 2009; Taubert et al. 2011; Varmavuo et al. 2013). However, these variances do not translate into relevant clinical differences regarding haematopoietic recovery. Taken together, the graft quality from poor mobilizers can be regarded equivalent compared to that from good mobilizers, regardless of the use of plerixafor.

Since CD34 is a marker for heterogeneous subgroups of cells, the composition of hematopoietic subpopulations as progenitor cells of the hematopoietic system might contribute to a better understanding of the graft quality (Roug et al. 2014; Saraceni et al. 2015; Dmytrus et al. 2016; Heuer et al. 2023). However, based on the cur-

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rently published data, no final conclusion can be drawn and further investigations are warranted to determine the potential effects of autograft cell subsets on the patients' clinical outcomes. As delineated in an EBMT position statement, determination of cell subsets other than $CD34^+$ cells is not routinely performed in clinical practice but only in clinical trials (Mohty et al. 2014). Accordingly, assessment of tumour cell contamination is usually not part of the clinical routine but can be of interest in clinical trials.

20.2 HSC Cryopreservation

HSC should be processed and stored in accordance with the respective Medical Council, responsible local and overarching authorities as well as scientific society's guidelines (e.g. EU: Guideline 2004/23/EG 2004; Guideline 2006/17/EG 2006; EU-GMP-Guideline 2012; Heuer et al. 2023).

If necessary, collected cells can be stored for a maximum of up to 72 h at 2–6 °C before cryopreservation. However, cryopreservation within 48 h or less is recommended to maintain an optimal viability of the cells. In the case of storage for >24 h prior to cryopreservation, the maximum nucleated cell (NC) concentration should not exceed 2×10^8/mL.

For cryopreservation, a number of different protocols are used worldwide. Usually, the maximum accepted NC concentration is $\leq 4 \times 10^8$/mL. If necessary, PBSC products can be diluted with autologous plasma, human albumin, or commercial resuspension medium. Increasing the cell concentration by volume depletion minimizes the number of cryostored bags needed, but the upper limit of the NC concentration needs to be considered. The final product includes 5–10% dimethyl sulfoxide (DMSO) as a cryoprotectant and 0.05–0.25 mL of ACD-A stabilizer solution per ml of transplant. Freezing at a controlled rate of 1–2 °C per minute is recommended. Cells need to be stored in the vapour phase nitrogen at a temperature of ≤−140 °C. Cross-contamination while preparing and storing the cells must be prevented by taking appropriate measures.

At the time of auto-HCT, cryopreserved bags must be thawed at the site of transplantation and PBSCs should be reinfused within a maximum time span of 20 min of thawing using standard transfusion filters in order to minimize the detrimental effect of DMSO upon HSC. Previous washing for purposes of DMSO depletion is not routinely performed, as the loss and damage of HSC are regarded as too high.

Several studies demonstrated that under these storage conditions, $CD34^+$ HSC remained viable for up to 19 years (Fernyhough et al. 2013; McCullough et al. 2010; Spurr et al. 2002). In addition, it could be demonstrated that the duration of cryostorage of the transplant had no impact on the haematologic reconstitution after transplantation (Lisenko et al. 2017).

20.3 HSC Quality Assessment

HSC product quality assessment needs to be performed at several time points during cell processing and storage. Volume measurement, enumeration of NC and red blood cells, and flow cytometry-based $CD34^+$ cell quantification should be performed directly after PBSC collection in accordance with the Stem Cell Enumeration Committee Guidelines of the International Society of Hematotherapy and Graft Engineering (ISHAGE) (Sutherland et al. 1996). A validated protocol and external quality control (e.g. the round robin test) is strongly recommended (Whitby et al. 2012).

Shortly before freezing, microbiological culture samples must be obtained. NC enumeration and NC viability measurement (e.g. by staining with trypan blue, 7-aminoactinomycin D [7-AAD] or propidium iodide) should be performed from aliquots of the final cell product after freezing and thawing. This viability testing is only valid for a defined and limited time span, often 2–5 years based on local guidelines, before it needs to be repeated prior to transplantation. Therefore, a sufficient number of reference samples should be prepared for each HSC product (the recommended minimum number is 3).

Target values need to be defined for the final product, mostly in accordance with local authorities. In most transplant centres in Europe, the following criteria are mandatory (together with additional criteria) for the release of an autologous transplant: NC concentration $\leq 4 \times 10^8$/mL, $CD34^+$ cell number $\geq 2 \times 10^6$/kg bw, red blood concentration ≤ 0.1 mL per mL of transplant, no microbial growth and minimum NC viability of >70% after freezing and thawing.

20.4 Collection of Reference Samples for Quality Control

Reference (or retention) samples for quality control must be taken and stored from the cell product. These samples allow the proof of quality and potency of the transplant in terms of sterility, purity, and viability of the cells. In the case of an allo-HCT, reference samples may also need to be collected from the donor, depending on the respective local legal requirements, to allow for a retrospective analysis in terms of serological testing.

Reference samples are prepared in parallel with the cell product and should be stored under the same cryoconditions until they are analysed. As a release criterion for an autologous stem cell transplant, a reference sample should be cryopreserved for >24 h under the identical conditions as the cell product before the viability of nucleated cells is analysed. Performing a clonogenic assay (e.g. colony-forming assay) from the reference samples can assess the haematopoietic potency of the cells. However, this is not regarded as a release criterion but should be performed for process validation or in the case of prolonged cryostorage of a transplant (>5 years).

The final cell product must be labelled in accordance with respective legal requirements. In order to transport a cryopreserved HSC product, a validated shipping container is required, preferably with continuous temperature monitoring. The treating physician is responsible for the application of the HSC transplant after evaluating its integrity and the accompanying documents.

Key Points

- Minimal number of $CD34^+$ cells is $\geq 2.0 \times 10^6$/kg bw for an auto-HCT and $\geq 4.0 \times 10^6$/kg bw for an Allo-HCT.
- Determination of HSC subsets is not routinely required, but may be part of clinical studies.
- Cryopreservation needs to be performed within 72 h, preferably <48 h.
- The maximum NC concentration in the cryostored transplant should be $\leq 4 \times 10^8$/mL.
- The final product includes 5–10% DMSO as a cryoprotectant and 0.05–0.25 mL of ACD-A stabilizer solution per ml of transplant.
- Freezing at a controlled rate of 1–2 °C per minute is recommended, and cells need to be stored in vapour phase nitrogen at a temperature of ≤ -140 °C.
- NC viability should be >70% after freezing and thawing.
- At the time of ABSCT, cryopreserved bags must be thawed and reinfused within a maximum of 20 min. of thawing.
- Reference samples for quality control must be prepared and cryostored in parallel and under identical conditions as the cellular product.

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